KAVITHA KANNAN



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Date of Birth: 1st July, 1994 ◇ Place of Birth: Chennai, India
Present location: Konstanz, Germany

PROFESSIONAL SUMMARY

I recently completed my Ph.D. in Biology, where I investigated honeybee behavior and neurobiology. Over the past five years, my research has spanned animal behavior, molecular biology, and quantitative data analysis, combining controlled laboratory experiments with fieldwork in remote environments. I have long been fascinated by how and why animals behave the way they do — how decisions are made, what sensory and social cues drive behavioral change, and the neural mechanisms that underlie these processes.

As I move forward, I am eager to broaden my perspective from individual-level mechanisms to large-scale ecological patterns, on exploring how environmental factors influence decision-making and shape evolutionary life histories, especially under the pressures of climate change. I also aspire to integrate my research background with biology education, science communication, and environmental policy to address conservation challenges on a broader scale.

EDUCATION

Doctor of Philosophy (Ph.D.), Biology, University of Konstanz, Germany — Jul 2025
Thesis: "The Defensive Behaviour of Honey Bees: Individuality, Social regulation and Neural mechanisms"

Master of Science (M.Sc.), EUCOR Joint Master's in Neuroscience,
Courses in Strasbourg, Freiburg and Basel, University of Strasbourg, France — Nov 2018
Thesis: "Expression analysis of differentially regulated genes of FoxP1, in the auditory perception of female zebra finches"

Bachelor of Science (B.Sc.), Triple major in Chemistry, Botany and Zoology,
St. Joseph's College of Science and Arts, Bangalore University, India — Jun 2015
Thesis: "Boldness trait and cognition in the Climbing Perch fish *Anabas testudineus*"

School, Indian School Certificate Board of Education (Science major),
The Valley School, Krishnamurti Foundation India, Bangalore, India — Jun 2012

EXPERIENCE

Doctoral Researcher / Research Associate — **Oct 2020 – Dec 2025**
AG Nouvian/AG Galizia, Department of Biology, University of Konstanz — *Konstanz, Germany*

- Investigated the **neuroethology** of **honey bee**, by focussing on their collective defensive behavior, chemical communication, individuality, social regulation, and neural mechanisms.
- Designed, implemented, and analyzed **lab-based behavioral assays** and **cellular biology methods**, including bee brain dissection, immunohistochemistry, and **confocal microscopy**. Developed **data analysis pipelines** for processing and visualizing complex biological datasets.
- Contributed to two **first-author publications** in reputed journals in the field.
- Conducted **fieldwork** in South Africa to collect behavioral data on African honey bee populations for comparative studies.
- Mentored one Master's student (6-week intensive VTK project) and one Bachelor's thesis student on a large video dataset for tracking bees in the stinging assay; supervised five HiWi students assisting in data collection.

Research Assistant **Jan 2019 - Oct 2020**
Lund Vision Group, Department of Biology, Lund University *Lund, Sweden*

- Conducted comparative neuroanatomy research of migratory and non-migratory moths, and analyzed large electrophysiological datasets from neuronal recordings in the Panamanian Sweat Bee (*Megalopta genalis*), creating **data visualizations** and **publication-ready figures**, for 2 peer-reviewed publications.
- Initiated **ethological experiments** on a novel model organism, the scorpion (*Heterometrus spinifer*), in the laboratory, which served as the basis for a competitive PhD grant proposal.

Research Internships and Academic Training **2013 - 2018**
Various institutes *India and Europe*

- **Master Thesis**, Language and Genetics Lab (Prof. Dr. Simon Fisher), Jan – Jun 2018
  Max Planck Institute for Psycholinguistics, *Nijmegen, Netherlands*
- **Bachelor Thesis**, Social Cognition Lab (Dr. VV Binoy), Nov – Feb 2015
  NIAS, Indian Institute of Science, *Bengaluru, India*
- **Field Research Intern**, May – Jun 2013
  Kaigal Conservation Center (Dr. Sudha Premnath), *Chittoor, India*

PUBLICATIONS

Kannan K, Galizia CG, Nouvian M. Consistency and individuality of honeybee stinging behaviour across time and social contexts. Royal Society Open Science. 2025 Jan 29; 12(1):241295.

Kannan K, Galizia CG, Nouvian M. Olfactory strategies in the defensive behaviour of insects. Insects. 2022 May 18; 13(5):470.

Adden A, Garcia Dominguez S, Kliem K, **Kannan K**, Yuvaraj JK, Raif T, Boronat-Garcia A, Arganda S, Talavera G, Kelber A, Heinze S. The evolution of lepidopteran brain morphology. Journal of Comparative Physiology A. 2025 Dec 22:1-24.

Honkanen A, Hensgen R, **Kannan K**, Adden A, Warrant E, Wcislo W, Heinze S. Parallel motion vision pathways in the brain of a tropical bee. Journal of Comparative Physiology A. 2023 Jul; 209(4):563-91.

OUTREACH ACTIVITIES

Science Communicator on Social Media Aug 2023
Instagram outreach focussing on bee-science and the daily life of a PhD. *Online Platform*

Science Communication Speaker Jul 2023
Public talk at the Munich Soapbox Science *Munich, Germany.*

Conference Organizing Committee Member May 2023
Neurobiology Doctoral Students Workshop (NeuroDoWo) *Konstanz, Germany.*

Youth Coordinator Dec 2013 – Mar 2015
"Neralu" Tree Festival *Bangalore, India.*

REFERENCES

- **Morgane Nouvian**, Neurobiology Group, Department of Biology, University of Konstanz, Germany. Email: morgane.nouvian@uni-konstanz.de. Phone: +49 7531 88 2115

- **Giovanni Galizia**, Neurobiology Group, Department of Biology, University of Konstanz, Germany. Email: giovanni.galizia@uni-konstanz.de. Phone: +49 7531 88 2238